Jatin Aggarwal

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EDUCATION

IILM University, Gurugram 2023 – 2027
Bachelor of Technology, Computer Science with specialization in AI and ML

SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL **Frameworks:** React.js, FastAPI, Node.js
Tools: AWS, Git, GitHub, GitLab, Docker, Linux, Postman **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Web Engineer Intern, Bluella (Cybersecurity and Infra), Gurugram May 2026 – July 2026

- Engineered and delivered **28+ production web deliverables** across healthcare, SaaS, and enterprise domains, including specialized healthcare portals, SaaS/product interfaces, responsive HTML email systems, and enterprise web interfaces.
- Built responsive front-end architectures using **HTML5, CSS3, and Vanilla JavaScript**, applying semantic HTML, CSS design tokens, Grid/Flexbox, fluid typography, theme systems, and DOM-based interactions.
- Implemented **SEO, accessibility, and performance optimizations**, including ARIA landmarks, structured metadata, responsive layouts, Core Web Vitals practices, and cross-device validation.

Chair, IILM GURUGRAM ACM Student Chapter January 2026 – Present

- Led the technical direction of the ACM Student Chapter, coordinating hands-on engineering initiatives, technical events, and student developer activities.
- Organized **Build → Pitch → Win**, a web application development marathon, and a full-scale ACM event featuring technical sessions, expert talks, and a technology-focused startup pitch competition.

Technical Workshop Lead, AISyncTech Club, IILM University, Gurugram 2025

- Led a hands-on technical workshop covering **SDLC workflows, virtualization, CLI tooling, and VM provisioning**.

PROJECTS

WeatherGPT — Multilingual Agro-Advisory & Emergency IVR Platform Live
FastAPI, React, PostgreSQL, Exotel Telephony, Sarvam AI, Google Gemini, Open-Meteo, Docker, AWS(EC2)

- Engineered a dual IVR telephony engine (Exotel and Web Simulator) with DTMF navigation, real-time voice barge-in, auto-retries, and SMS fallback for automated outbound disaster broadcasts and inbound advisory calls.
- Designed a location-aware agro-meteorological advisory engine evaluating **5 agronomic rules** covering irrigation, spraying, fungal disease risk, heat stress, and fertilizer timing, with multilingual voice/text processing in Hindi, Kannada, and English.
- Built and containerized a full-stack platform with a FastAPI asynchronous backend, React dashboard, and PostgreSQL for call tracking and alert management.

MariaDB Magics — Jupyter Extension for SQL, Analytics and Vector Search GitHub
MariaDB Python Hackathon 2025 — Top 59 / 4,000+ teams

- Built a custom Jupyter extension providing **mariadb, mariadb plot, mariadb time, and mariadb vector** magics for SQL execution, visualization, temporal analysis, and semantic vector search directly inside notebooks.
- Engineered modular Python packages integrating the MariaDB Python connector, sentence transformer embeddings, Matplotlib, and optimized query execution pipelines.
- Enabled an AI and database workflow for developers and analysts, reducing context switching, speeding exploratory analysis, and unlocking semantic search capabilities on MariaDB.

VoiceOps — Enterprise Voice AI Observability Engine GitHub Live
Python, FastAPI, Pipecat DAG, Gemini Live, Groq, Upstash Redis, PostgreSQL, Streamlit

- Architected a Voice AI observability platform using a **Hexagonal Architecture**, decoupling continuous **Pipecat audio DAGs** from the FastAPI ASGI event loop via `asyncio` to achieve $<$ **5 ms REST API dispatch times** without starving the web server.
- Engineered a **Deterministic Sidecar Interceptor** that halts LLM audio generation to enforce database invariants during tool calls, alongside a custom TTS adapter utilizing an in-memory **`ffmpeg` IPC pipe** to stream 16kHz PCM audio with **zero disk I/O overhead**.
- Designed a non-invasive telemetry processor using the **Transactional Outbox Pattern** (Upstash Redis to Supabase PostgreSQL) to completely decouple database writes from the audio stream, validating the system with **63/63 asynchronous tests passing**.